

A Hybrid Machine Learning and Rule-Based Decision Support System for Short-Term Trading: A Case Study on XAUUSD

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Abstract

Financial markets, characterized by high liquidity and rapid capital flows, require reliable predictive tools to support trading decisions and risk management. This study evaluates several machine learning algorithms to identify a robust short-term price forecasting model and, based on these findings, develops a rule-based decision-support system for algorithmic trading. Using XGBoost with tailored feature engineering, the proposed model forecasts five-minute gold (XAU/USD) price movements over a one-hour horizon, achieving a root mean square error (RMSE) of 3.49 and a mean absolute percentage error (MAPE) of 0.11%. When operationalized as a complete trading strategy governed by a fixed-dollar-risk position-sizing rule, the system produced a cumulative net return of 130.62% (\$13,062.38 profit on a \$10,000 initial capital) over the January 2021 - October 2025 period, with Monte Carlo simulation indicating a mean annualized return of approximately 30.7% and an 89% probability of annual profitability. These results highlight the model's effectiveness for data-driven investment and automated trading applications.

Keywords: *algorithmic trading, machine learning, financial markets, price prediction.*

1. Introduction

The rapid expansion of artificial intelligence (AI) within financial markets, coupled with the growing prevalence of algorithmic trading, has intensified the need for reliable data-driven forecasting and execution systems. As market structures become increasingly complex and trading speeds accelerate, accurate short-term price prediction is essential for effective risk management and sustained profitability (Enkhbayar & Ślepaczuk, 2025). This study focuses on the spot price of gold, a highly liquid commodity with a stable microstructure and reduced susceptibility to abrupt shocks compared with lower-liquidity assets such as many cryptocurrencies. Although algorithmic strategies are theoretically market-agnostic, empirical performance is heavily

influenced by liquidity regimes, and highly liquid instruments consistently yield more reliable predictive signals.

Grounded in the Efficient Market Hypothesis (Fama, 1970) and its technical-analysis interpretation (Murphy, 1999), this research restricts inputs to price-derived features and classical technical indicators, excluding fundamental data, news sentiment, and alternative datasets, on the premise that publicly available information is rapidly incorporated into price action.

Despite substantial advances in machine learning and deep learning for financial prediction between 2021 and 2025, reported results remain heterogeneous, and few studies systematically integrate high-quality forecasts into executable trading logic with rigorous risk controls (Dakalbab et al., 2025; Chan, 2013). Existing gaps include (i) limited attention to feature-engineering practices tailored to non-stationary financial series, and (ii) reliance on simple historical back testing without stress-testing under randomized scenarios.

This paper addresses these shortcomings by proposing and empirically evaluating a complete end-to-end algorithmic trading system for gold spot prices. The framework comprises (a) a machine learning ensemble for ultra-short-term directional forecasting, (b) feature-engineering enhancements and rule-based filters to improve signal robustness, (c) a fixed-dollar-risk position-sizing rule that keeps monetary risk per trade constant regardless of volatility, and (d) comprehensive risk-management overlays. Model performance is assessed through walk-forward back testing and extensive Monte Carlo simulations (Bajgrowicz & Scaillet, 2012; Mendes & Lavrado, 2017) to distinguish genuine edge from overfitting and to quantify tail-risk exposure.

The primary contribution is a unified, reproducible pipeline that translates predictive signals into risk-aware, executable orders while demonstrating consistent profitability and drawdown control in a highly liquid market. The results offer practical guidance for practitioners and establish a benchmark for future hybrid predictive-algorithmic systems.

2. Literature Review

The field of algorithmic trading and financial price prediction has grown rapidly, with roughly 200-500 articles published annually over the past decade in databases such as ScienceDirect. A critical examination of this literature reveals persistent challenges. First, many published models perform well in controlled, simulated environments but fail to translate into consistent real-world profitability (Chan, 2013). Second, genuinely profitable strategies are rarely disclosed publicly, since public adoption would arbitrage away their edge (Harris, 2015). Consequently, most openly available strategies either contain undisclosed risks, overfit historical data, or remain conceptual prototypes requiring substantial customization before live deployment.

This gap is particularly visible on the MetaTrader platform, the most widely used retail trading environment. A systematic search of ScienceDirect (1990-2025) identified only four articles explicitly implementing multi-criteria, decision-making-based algorithmic systems on MetaTrader, despite the platform's fidelity to live market microstructure.

Recent contributions (2021-2025) have advanced methodological sophistication without consistently delivering operationally viable systems. Hybrid sentiment-technical and deep-learning models for next-day or intraday price prediction (Jing et al., 2021; Bose et al., 2021; Hoseinzade & Haratizadeh, 2021; Bansal & Goyal, 2022) report improved accuracy but retain prediction errors too large for standalone trading, or omit strong ensemble baselines such as XGBoost and random forests. Sentiment-augmented and optimized deep-learning approaches (Cohen, 2022; Gülmez, 2023) improve directional accuracy but remain untested in high-volatility regimes, while limit-order-book-based models (Bilokon & Qiu, 2023; Yin & Wong, 2023) achieve strong simulated returns using data rarely available outside institutional or centralized venues.

Reinforcement-learning approaches have also gained traction, mimicking professional traders or targeting reversal signals during market stress (Da & Yeong, 2024; Song, 2024), but typically suffer from excessive drawdowns, small sample sizes, or poor cross-market transferability. CNN-BiLSTM-attention and LSTM-GARCH hybrids (Abhay & Vishwakarma, 2024; Xu et al., 2024) reduce forecasting error but are often restricted to single-step horizons unsuitable for most trading styles, and sentiment-augmented portfolio scoring methods (Ricchiuti & Giancarlo, 2024) achieve short-term outperformance accompanied by severe drawdowns (40-60%). Further studies (Lu & Xu, 2024; Ghadiri & Hajizadeh, 2025; Dakalbab et al., 2025; Qin et al., 2025) illustrate recurring themes: sophisticated architectures deliver statistical improvements yet routinely lack live-market validation, carry high overfitting risk, or report risk-adjusted metrics undermined by large drawdowns.

Several 2025 studies (Enkhbayar & Ślepaczuk, 2025; Hiruta & Senoguchi, 2025; Huang et al., 2025) reaffirm increasing market efficiency in forex and equities, reinforcing the challenge of sustainable alpha. Across the literature, XGBoost frequently emerges as a robust baseline, often outperforming more complex deep-learning alternatives at medium horizons, underscoring that architectural novelty does not guarantee practical superiority.

In summary, the literature reveals a systematic failure to bridge predictive modeling and risk-controlled, executable systems tested under authentic market conditions. The present research addresses this void by developing and rigorously evaluating a complete, MetaTrader-native trading framework for gold spot prices that integrates machine learning forecasting, rule-based filters, and comprehensive risk management, delivering verifiable, real-market-ready performance where prior work has fallen short.

3. Research Gap and Positioning of the Present Study

The methodological landscape of recent algorithmic trading and price-prediction studies (2019-2025) spans sentiment-augmented deep neural networks, reinforcement learning, hybrid LSTM-GARCH, attention-based transformers, and fuzzy multi-criteria systems. Yet this literature shows a conspicuous void: the near-total absence of comprehensive real-market validation metrics (e.g., maximum drawdown, Profit Factor, Sharpe ratio, annualized return under transaction costs, and Monte Carlo stress testing) and live execution frameworks.

This deficiency underscores the core research gap: an enduring disconnect between theoretical sophistication and operational viability. Although contemporary models frequently achieve impressive statistical benchmarks offline, they rarely progress to live deployment on production-grade platforms such as MetaTrader, where slippage, latency, microstructure noise, and regime shifts impose binding constraints (Chan, 2013; Harris, 2015). Moreover, risk-adjusted performance indicators are either omitted or incompletely reported, rendering claims of profitability inconclusive under realistic trading conditions.

A recurring observation is the under-exploitation of gradient boosting frameworks such as XGBoost. Despite its demonstrated robustness and frequent outperformance of deep-learning alternatives at medium horizons (Enkhbayar & Ślepaczuk, 2025; Hiruta & Senoguchi, 2025; Géron, 2022), existing applications treat XGBoost as a static predictor rather than an adaptive, time-series-aware engine capable of real-time signal generation.

The present study addresses these shortcomings by proposing a fully executable algorithmic trading system for gold spot prices centered on a temporally engineered XGBoost ensemble. Unlike prior work, the framework (i) incorporates recursive and lagged feature construction to capture dynamic market regimes, (ii) embeds a fixed-dollar-risk position-sizing rule together with rule-based confirmation filters for signal refinement, (iii) implements comprehensive risk overlays (position limits, volatility-adjusted scaling, and trailing protection), and (iv) undergoes rigorous evaluation through walk-forward back testing and extensive Monte Carlo simulation. By translating a conventionally "static" algorithm into a live, risk-aware trading system, this research establishes a reproducible benchmark for converting predictive accuracy into sustained, risk-adjusted profitability in a highly liquid yet volatile asset class.

4. Methodology

This study adopts a systematic, end-to-end quantitative research design integrating theoretical foundations with rigorous empirical validation. The methodology unfolds in four phases: (i) theoretical framework construction, (ii) data preparation and temporally aware feature engineering, (iii) predictive model development and hyperparameter optimization, and (iv)

integration into a risk-controlled executable trading strategy evaluated under realistic market conditions.

The predictive core employs an enhanced XGBoost ensemble (Chen & Guestrin, 2016), selected for its proven handling of non-stationary financial time series, resistance to overfitting through built-in L1/L2 regularization, and computational scalability. The implementation incorporates recursive and lagged technical indicators, volatility-regime features, and rolling-window adaptations to capture evolving market dynamics while preserving strict chronological integrity (no look-ahead bias). Historical spot gold prices (XAU/USD) from January 2021 to October 2025 are sourced at multiple resolutions (M15, H1, and daily) from a regulated broker feed, preprocessed for missing values and outlier synchronization, and split using a purged walk-forward scheme to emulate live deployment.

Data acquisition and preprocessing were carried out through a dedicated, resource-constrained, bias-aware data pipeline developed for this purpose, which sources prices directly from live MetaTrader 5 broker execution feeds rather than retrospective aggregator data, and handles gap-filling, calendar reconstruction, indicator warm-up periods, and multi-symbol merging as explicit, auditable, strictly backward-looking procedures (Masoumian, 2026). This pipeline was independently validated for the absence of look-ahead bias and reproducibility (via deterministic MD5-hash checks) prior to its use in the present study, and its implementation is publicly available on GitHub, providing full auditability of the data-preparation steps underlying the results reported here.

Hyperparameter optimization is conducted via grid search over key hyperparameters (`n_estimators`, `max_depth`, and `learning_rate`), using purged cross-validation on the training period with folds anchored to expanding training windows to guard against regime shifts. Directional forecasts are transformed into executable signals through a multi-layer filtering pipeline that combines probability thresholds, a fixed-dollar-risk position-sizing rule, rule-based confirmation filters (e.g., trend alignment, volatility caps), and dynamic risk overlays (trailing stops, volatility-adjusted exposure, and maximum drawdown breakers).

Performance is assessed using a comprehensive hierarchy of metrics: out-of-sample predictive accuracy (MAE, RMSE, directional hit rate), trading-oriented financial indicators (annualized return, Sharpe and Sortino ratios, Profit Factor, maximum drawdown, Calmar ratio), and statistical robustness validated through Monte Carlo permutation tests and stress scenarios. The system is fully implemented as a native Expert Advisor on the MetaTrader 5 platform, ensuring fidelity to real-market microstructure, including variable spreads, swap fees, and execution latency.

5. Dataset and Features

The dataset consists of over 300,000 five-minute candlesticks for spot gold (XAU/USD) available in kaggle¹, covering January 2021 to October 2025, sourced from a regulated broker to ensure high-fidelity tick data. Each record contains standard OHLCV fields augmented with engineered features spanning established technical indicators (e.g., SMA/EMA, RSI, MACD, Parabolic SAR, Bollinger Bands, ATR, Stochastic Oscillator, and pivot-point levels) and proprietary features developed for this study to capture contextual and regime-specific price behavior (e.g., distance and directional bias relative to pivot levels, candle body/shadow ratios normalized by volatility, fractal-volume filters, and higher-timeframe trend alignment signals). An initial, larger feature set was pruned using Random Forest feature-importance ranking to remove low-contribution and redundant variables, yielding a final set of 28 features that keeps the model lightweight while preserving predictive power. The final feature list is presented in Table 1. All features are computed recursively using only past information to eliminate look-ahead bias and ensure full live-tradability.

Table 1. Features Description

Row	Feature / Attribute	Description
1	date/time	Record the time of each candle (for sorting only, not used in the model)
2	open, high, low, close	The four main prices of the candle (the basic input of the model)
3	tick_volume	The number of activated orders per candle
4	spread	The difference between the bid and ask prices (liquidity indicator)
5	ATR	The average true range to measure market volatility
6	MA_Close	The 5-window moving average based on the closing price
7	MA_Open	The 5-window moving average based on the opening price
8	MA_Trend	The 15-candle moving average to identify the short-term trend
9	pivot	The pivot level with a window of 96 (equivalent to 8 hours)
10	RSI_raw	The raw value of the Relative Strength Index (RSI)
11	MA_cross	The state of the intersection of the moving averages
12	diff_pivot	The distance of the price from the pivot level
13	higherhigh	The highest high in the last 5 candles
14	lowerlow	The lowest low in the last 5 candles
15	leight	The distance to the low of the candles
16	lower_third	Lower third of the candle
17	upper_third	Upper third of the candle
18	fractal	Fractal indicator (reversal detection)

¹ <https://doi.org/10.34740/kaggle/dsv/18844088>

Row	Feature / Attribute	Description
19	PSAR	Parabolic SAR indicator, trend direction
20	MACD	MACD indicator, exponential moving averages
21	RSI	Discrete version of RSI_raw (above 70: -1; below 30: +1; else 0)
22	TN	Price position relative to MA_Trend
23	MA	Cross of two moving averages (1: bullish, -1: bearish, 0: neutral)
24	CA	Candle's closing position relative to the previous candle and its inner range
25	FV	Fractal indicator with volume filter (1: low, -1: high)
26	PV	Price position relative to pivot (above: -1, below: 1)
27	PVT	Distance of price from pivot based on ATR (less than 2: 1, more than 2: 0)
28	Decision	Combined binary signals, the final decision indicator (buy/sell/wait)

Samples are generated with a stride of one candle, preserving chronological order and natural market dynamics. The dataset is split temporally: 2021-2024 for training and hyperparameter tuning via purged walk-forward validation, and the full year 2025 is reserved as a pure out-of-time test set. This partitioning reflects recent market regimes dominated by algorithmic activity while providing sufficient data volume and recency for robust generalization under live conditions. Optimized hyperparameters are reported in Table 2.

6. Time-Series Configuration and Data Partitioning

The model employs a sliding-window approach with a fixed 12-candle lookback (equivalent to 60 minutes of past data in the 5-minute timeframe) and predicts the entire sequence of the next 12 closing prices (60-minute horizon). For each time t , all features are computed exclusively from the interval $[t-11, t]$, while the target vector y_t comprises closing prices from $[t+1, t+12]$. This fixed temporal gap fully eliminates look-ahead bias and ensures strict out-of-sample validity.

7. Hyperparameter Optimization

The XGBoost model was tuned using grid search over three key hyperparameters - `n_estimators`, `max_depth`, and `learning_rate` - with separate ranges selected based on preliminary experiments and common practices for financial time-series tasks. The optimized values, identified through purged cross-validation on the training period to minimize overfitting, are reported in Table 2.

Table 2. Hyperparameter tuning results for the predictive XGBoost model

#	Hyperparameter	Lower bound	Upper bound	Step	Optimal value
1	<code>n_estimators</code>	100	250	50	250
2	<code>max_depth</code>	5	11	3	11

#	Hyperparameter	Lower bound	Upper bound	Step	Optimal value
3	learning_rate	0.01	0.07	0.02	0.07

The next section evaluates the forecasting and trading performance of the final model configured with these hyperparameters.

8. Predictive Performance of the Price Forecasting Model

To evaluate the performance of the proposed XGBoost-based model, predicted prices were compared against actual values across the training and test datasets to assess the model's generalization capability and the alignment of predictions with actual market trends throughout the study period.

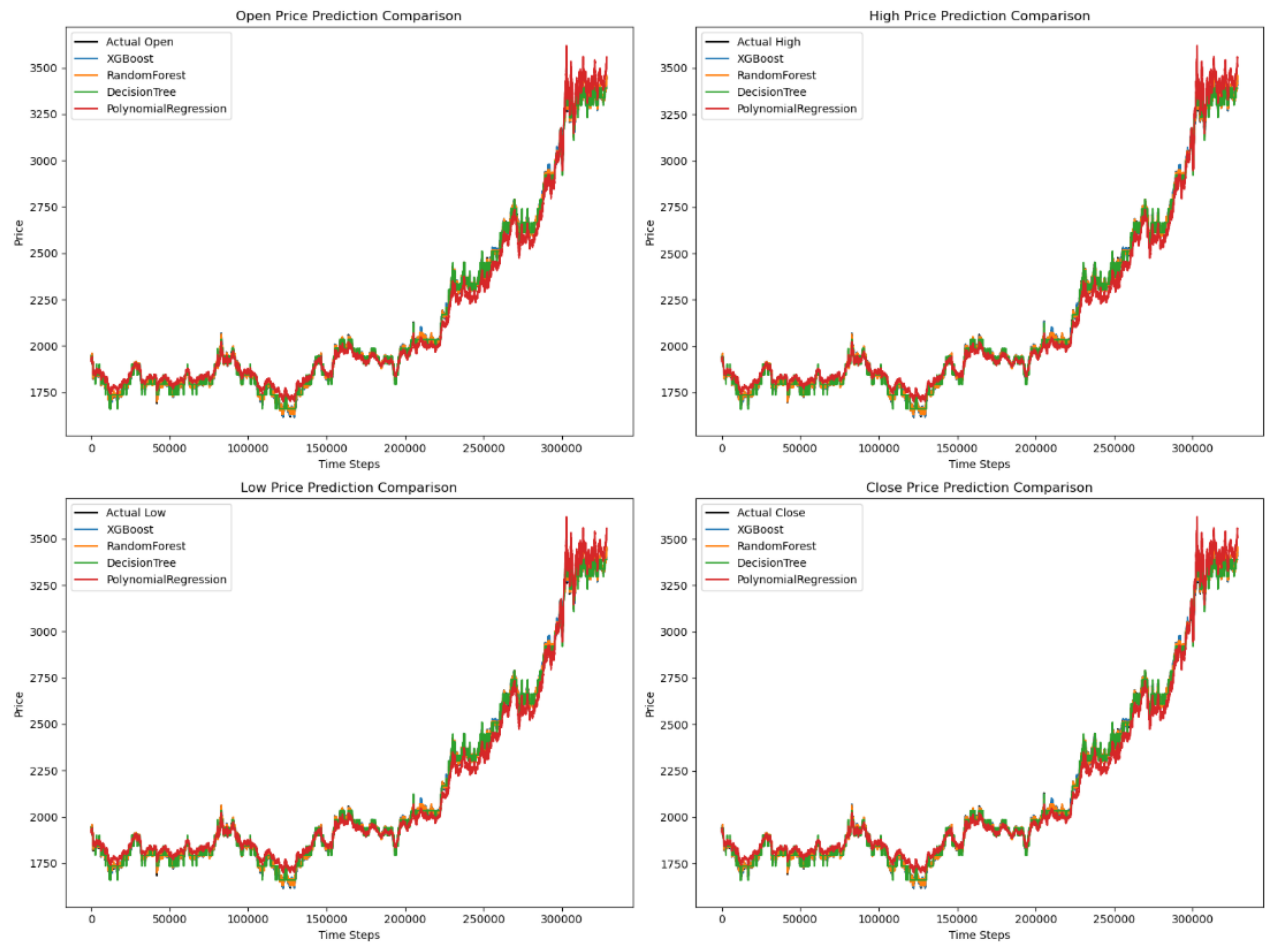


Figure 1. Comparison of predictive performance across the full dataset used

Figure 1 displays predictions of the four main prices (High, Low, Open, and Close) generated by the XGBoost, Random Forest, Decision Tree, and Polynomial Regression algorithms; the vertical axis represents price and the horizontal axis represents the time index. The XGBoost model tracks price changes more accurately and consistently than the other algorithms.

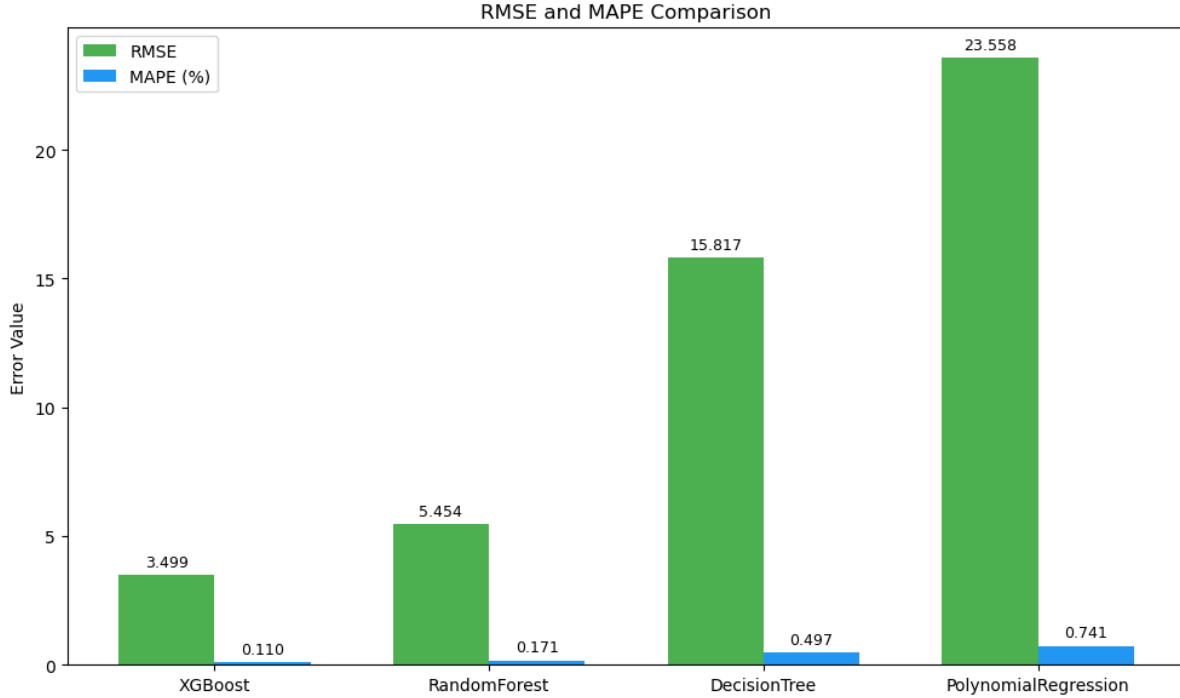


Figure 2. Performance evaluation metrics across algorithms

As shown in Figure 2, the error metrics confirm the satisfactory predictive accuracy of the proposed XGBoost model. The mean absolute percentage error (MAPE) is 0.11%, and the root mean square error (RMSE) is 3.49 (approximately 349 pips in XAU/USD). Given that typical 60-minute price fluctuations in spot gold during New York trading hours range between 250 and 410 pips across twelve five-minute candles, this level of error is considered acceptable for integration into a live trading system; generated signals nonetheless require additional rule-based validation and risk filtering before execution. The overlaid predictions in Figure 1 are produced using the final model, trained on the 2021-2024 period and evaluated across both the training window and the full 2025 out-of-time test set.

To further assess the model's generalization ability and the absence of overfitting under real-market conditions, predictions were also generated on a pure forward-looking out-of-sample period acquired after the initial dataset download and model finalization. As shown in Figure 3, performance on this unseen segment improved further, with MAPE decreasing to 0.06% and RMSE to 2.83 (approximately 283 pips), confirming the model's robust capability for reliable near-term forecasting (up to a 60-minute horizon) in live trading environments.

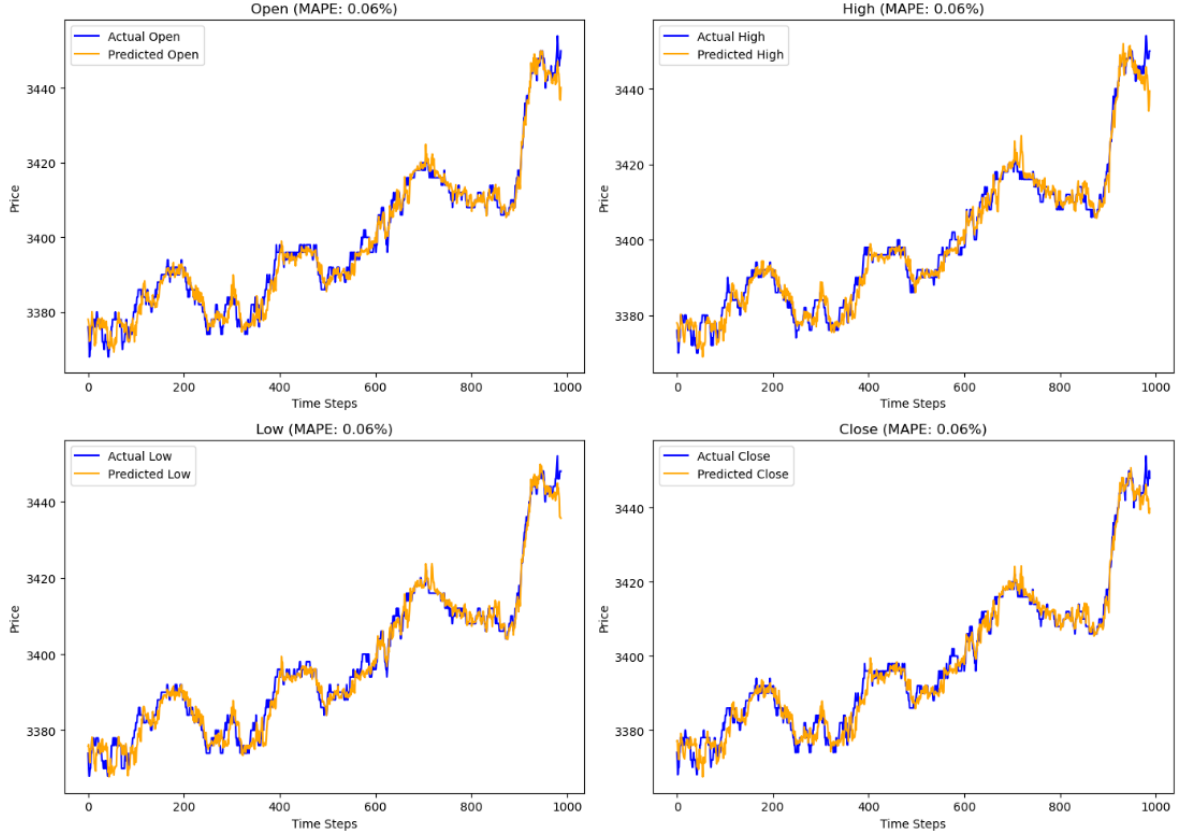


Figure 3. Out-of-sample predictive performance of the forecasting model

9. Trading Strategy

The proposed trading strategy integrates directional forecasts from the optimized XGBoost model with a multi-layer rule-based decision framework to generate executable buy, sell, or hold signals on the 5-minute timeframe. Predicted OHLC values for the next 12 candles are averaged to produce four proprietary indicators: MPO (mean predicted open), MPH (mean predicted high), MPL (mean predicted low), and MPC (mean predicted close). A normalized deviation score is computed as:

$$Score = (MPC - Current\ Close) / ATR(14)$$

A long signal is issued if $Score > \theta \times ATR(14)$, and a short signal if $Score < -\theta \times ATR(14)$, where θ is an empirically calibrated threshold. Additional confirmatory filters are applied using a three-row binary matrix incorporating the current and two prior candles across nine indicators (TN, MA, CA, FV, PV, PVT, RSI, PSAR, MACD), along with short-term trend alignment (moving averages and Parabolic SAR) and price position relative to pivot levels. Entry is prohibited during extreme candle ranges or when aggregate filter disagreement exceeds a critical level, while early-exit rules trigger on adverse trend reversals.

9.1 Rule-Based Filters

A rule-based filtering mechanism enhances the interpretability and practical applicability of the trading system. The filter is applied to a three-row rolling matrix derived from the dataset, where each row corresponds to the feature set of consecutive time steps, including Trend (TN), Moving Average signals (MA), Candlestick Analysis (CA), Fibonacci Values (FV), Pivot Points (PV), Price Volume Trend (PVT), Decision signals, RSI, Parabolic SAR (PSAR), MACD, and other price-based metrics, allowing the system to capture temporal dependencies and cross-indicator relationships simultaneously.

Opening rules are designed to prevent extreme or unreliable trades: trades are suppressed if the absolute difference between the open and close prices exceeds a predefined threshold, or if the aggregated decision indicator reaches extreme values. Buy and sell signals combine momentum and volatility indicators adjusted for ATR, together with consistency checks across short-term indicator trends, pivot proximity, and alignment between technical and price-based indicators, ensuring that signals are supported by multiple independent factors.

Early-closing rules reduce exposure to unfavorable market movements: positions are closed when key trend or candlestick indicators signal a reversal, or when the system detects potential overextension in the current position, enforcing a conservative exit strategy and mitigating drawdowns. This framework ensures signals generated by the machine learning predictions are filtered through well-defined, interpretable conditions, avoiding overreliance on black-box predictions - particularly important for short-term trading in high-liquidity instruments such as XAUUSD.

9.2 Entry Pricing and Position Sizing

Entry occurs at the closing price of the candle that generates a valid signal, eliminating look-ahead bias. The stop-loss for long positions is placed at the lowest low of the prior five candles; for short positions, at the highest high of the same window. Position size is governed by a fixed-dollar-risk sizing rule: a constant monetary amount (\$50 per trade) is risked irrespective of market volatility, with position size calculated from the distance between entry and stop-loss so that realized risk per trade remains consistent.

9.3 Risk Management

Risk control operates at multiple levels. Per-trade risk is capped by the fixed-dollar stop-loss and position-sizing rule described above. Daily profit/loss circuit breakers halt trading if equity deviation exceeds predefined multiples of historical daily volatility, protecting against black-swan events and regime shifts. A session filter, derived from back-test analysis, disables trading during historically underperforming hours (typically low-liquidity Asian sessions). Combined, these mechanisms - dynamic stop-loss, fixed-dollar risk, daily loss limits, and time-based filtering -

provide comprehensive capital protection and enhance the strategy's robustness in live market conditions.

9.4 Final Algorithm and Execution Logic

The deployed trading system employs a hybrid architecture that tightly couples the XGBoost directional forecasts with an optimized ensemble of rule-based filters, ensuring both predictive power and operational robustness.

Filter selection followed a structured forward-feature-selection process inspired by genetic algorithms, executed manually through exhaustive back testing. Binary and ternary-valued decision features (taking values in $\{-1, 0, +1\}$) were first evaluated individually via isolated trading simulations; features failing to generate statistically significant positive expectancy were discarded. Surviving candidates were then systematically combined in pairwise and triplet configurations, with performance ranked by net profitability, win rate, and risk-adjusted metrics (Sharpe ratio, Profit Factor, maximum drawdown). The top-performing three to four rules were retained as primary confirmatory filters and applied conjunctively to every candidate signal produced by the XGBoost module.

A subset of high-response pairwise rules, characterized by faster activation, was dedicated to dynamic exit management. Upon detection of a weak counter-directional signal, these rules prohibit new entries in the opposite direction while immediately closing existing positions. This asymmetric early-exit mechanism consistently truncates losing trades before the predefined stop-loss is reached, resulting in realized average losses substantially below the initial risk per trade and yielding the observed 1.5:1 reward-to-risk asymmetry.

The complete execution cycle, repeated at the close of every 5-minute candle, proceeds as follows:

9.4.1. Real-time OHLCV data are retrieved from the broker and subjected to on-the-fly preprocessing and recursive feature engineering.

9.4.2. The optimized XGBoost ensemble generates probabilistic forecasts for the next 12 candles, which are aggregated into the four mean predicted levels (MPO, MPH, MPL, MPC).

9.4.3. A normalized deviation score is computed relative to the 14-period ATR; if the absolute score exceeds the calibrated threshold ($\theta = 0.5$), a preliminary directional signal is triggered.

9.4.4. The candidate signal undergoes mandatory validation against the primary rule-based filter ensemble; only signals satisfying all active rules proceed.

9.4.5. Upon full confirmation, entry price (current close), stop-loss (80% of the five-candle swing extreme), take-profit ($1.5 \times$ risked distance), and the fixed-dollar-risk position size are calculated instantaneously.

9.4.6. The corresponding market order is transmitted to the broker's execution engine via the MetaTrader 5 API.

Concurrent positions are prohibited, and all risk overlays (daily loss/profit caps, session filtering, and early-exit logic) remain enforced throughout. This deterministic, low-latency pipeline - integrating state-of-the-art machine learning with transparent, interpretable rules - delivers a fully automated, live-tradable system that consistently translates predictive edge into risk-controlled profitability across varying market regimes.

9.5 Parameter Configuration and Optimization

Parameter selection is critical to the trading system's profitability and robustness. Fixed technical-indicator periods, aligned with the rule-based logic and kept constant throughout, are summarized in Table 3.

Table 3. Fixed technical indicator parameters

#	Parameter	Value
1	Moving average window on Open price	5
2	Moving average window on Close price	5
3	Moving average window on Pivot levels	96
4	Trend-detection moving average window	15
5	RSI & ATR period	15
6	Parabolic SAR step	0.2
7	Parabolic SAR maximum step	0.8
8	MACD fast period	10
9	MACD slow period	11
10	MACD signal period	8

An iterative forward-selection process, analogous to a genetic algorithm, was used to optimize the confirmatory rule-based filters. Combinations of individual, pairwise, and triplet rules were evaluated sequentially based on historical performance. Filters incorporating RSI and Parabolic SAR were ultimately discarded due to consistent degradation in risk-adjusted metrics and are excluded from the final trading system.

Key executable parameters - signal threshold, reward-to-risk ratio, stop-loss buffer, daily profit/loss caps, and active trading hours - were optimized via walk-forward analysis on the 2021-2024 period and validated out-of-sample in 2025. Optimized values are reported in Table 4.

Table 4. Optimized trading system parameters

#	Parameter	Lower bound	Upper bound	Optimal value
1	Signal issuance threshold (θ)	0.5	1.0	0.5
2	Reward-to-risk ratio	1.0	3.0	1.5
3	Stop-loss buffer (% of swing)	0.8	0.8	0.8
4	Daily loss limit (USD)	-600	-200	-200
5	Daily profit target (USD)	400	800	400
6	Active trading window (UTC+2)	01:00	23:00	8:00-20:00

The signal threshold governs the minimum normalized deviation (relative to ATR) required to trigger a trade; lower values increase trade frequency while raising the risk of false signals. The reward-to-risk ratio determines take-profit placement and reflects the expected profitability per unit of risk. The stop-loss buffer of 0.8 fixes the stop at 80% of the recent five-candle swing, achieving an optimal balance between win rate and average profit magnitude. Daily circuit breakers and the restricted high-liquidity trading window (London/New York overlap) serve as macro risk controls, preventing capital erosion during anomalous regimes and low-volume periods.

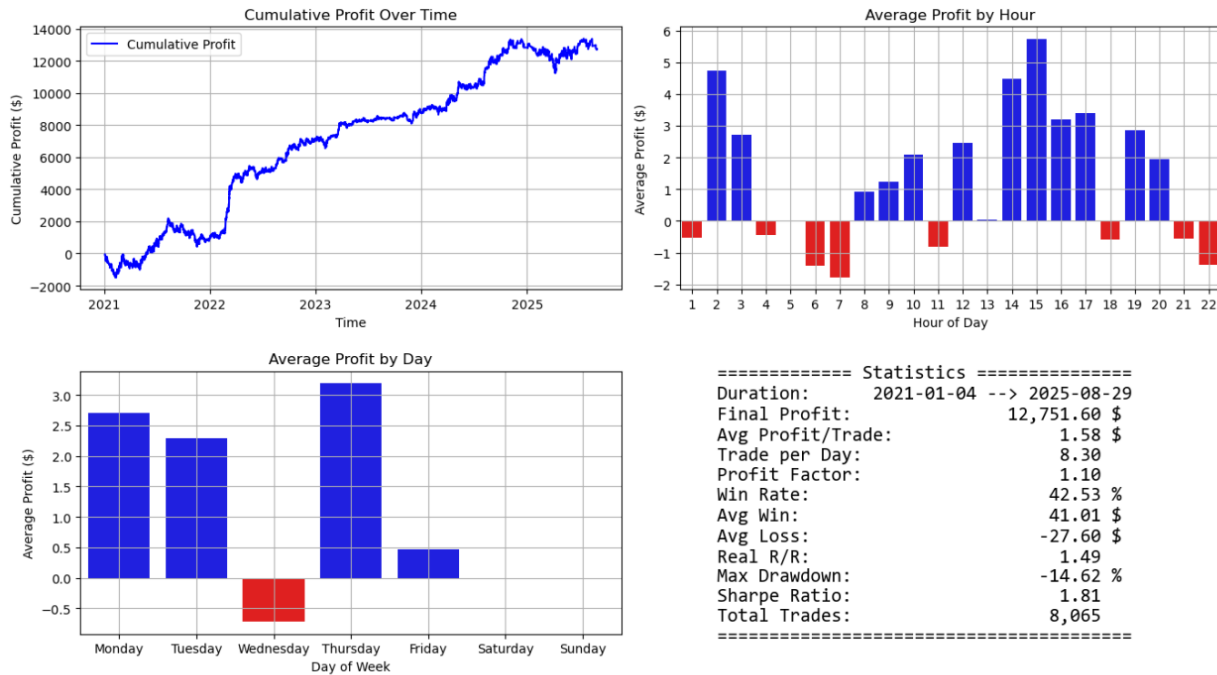


Figure 4. Active trading window of the basic system

Figure 4 illustrates the intraday trading window during which the algorithm is active and permitted to open positions. The optimal interval - determined by maximizing average hourly profitability across the historical back test - spans 08:00 to 20:00 (UTC+2), corresponding primarily to the

London and early New York sessions where liquidity and directional efficiency are highest for spot gold.

9.6 Final System Performance

In the final evaluation, the complete dataset spanning January 2021 to October 2025 was used in a unified walk-forward framework for training, parameter optimization, and out-of-sample testing. Simulations assumed an initial capital of \$10,000 and the fixed-dollar-risk rule of \$50 per trade.

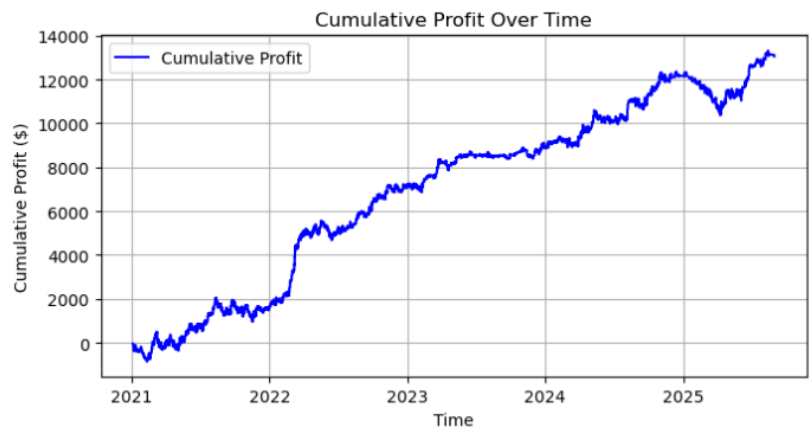


Figure 5. Cumulative profit trajectory of the final trading system

Figure 5 illustrates the cumulative net profit curve, which starts at zero and reaches \$13,062.38 by the end of the period - a total return of 130.62% over approximately five years (annualized $\approx 18.5\%$). A pronounced acceleration in equity growth is evident during early 2022, attributable to heightened volatility in the gold market triggered by the Russia-Ukraine conflict. The system consistently exhibits superior performance during high-volatility regimes, particularly at the openings of the Tokyo and London sessions and throughout New York trading hours.

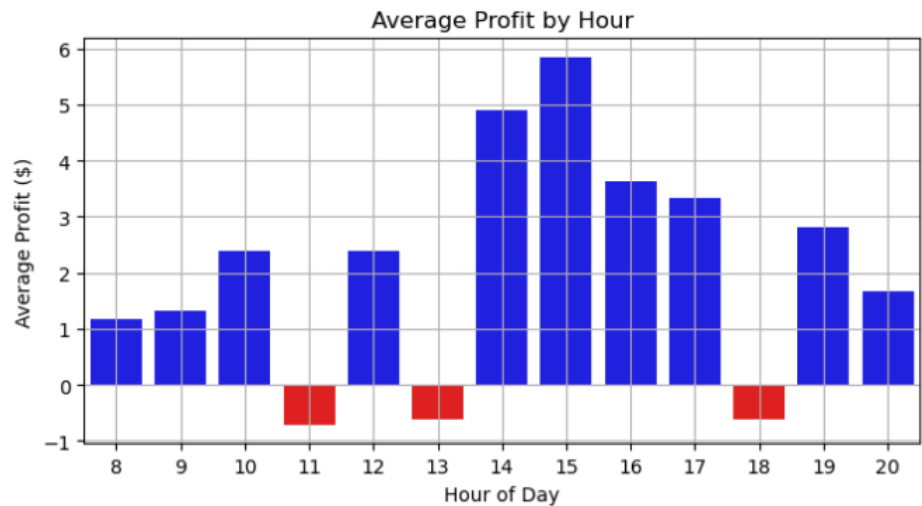


Figure 6. Average hourly profitability of the final system

Figure 6 presents the mean profit per hour (UTC+2) across the trading day. Peak performance occurs during the London-New York overlap (14:00-15:00), followed by the broader European-American session, confirming the model's efficacy in environments characterized by elevated liquidity and directional momentum.

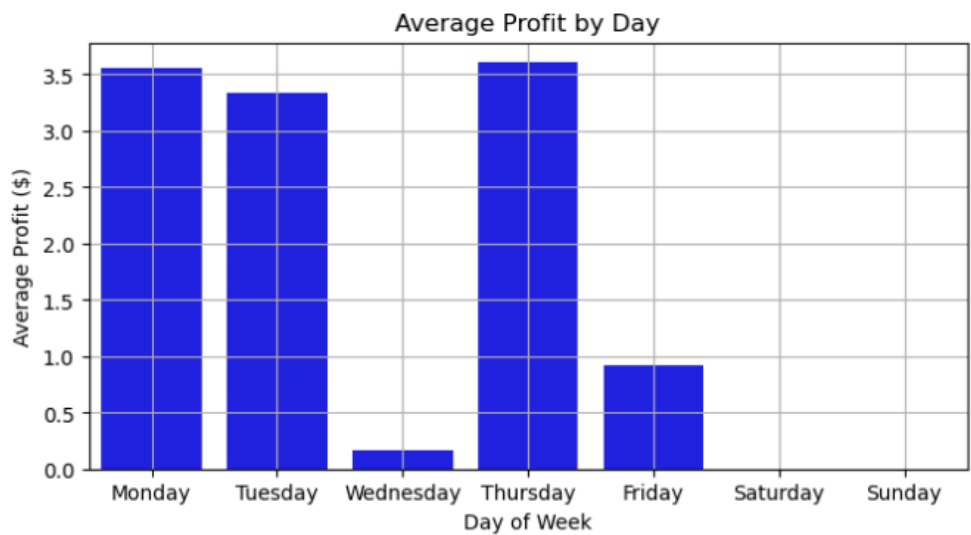


Figure 7. Average daily profitability by weekday

The daily return histogram in Figure 7 reveals significantly higher average profits on Mondays, Tuesdays, and Thursdays ($\approx$ \$3.50 per day), coinciding with elevated intraday volatility and sentiment-driven moves that align closely with the system's logic. Conversely, Wednesdays exhibit the weakest results ($<$ \$0.25), reflecting subdued ranging conditions, while Fridays - despite occasional volatility - suffer from irregular end-of-week position squaring that reduces pattern predictability.

```
===== Statistics =====
Duration:      2021-01-04 --> 2025-08-29
Final Profit:      13,062.38 $
Avg Profit/Trade:    2.30 $
Trade per Day:      6.24
Profit Factor:      1.14
Win Rate:          43.40 %
Avg Win:           43.88 $
Avg Loss:          -29.57 $
Real R/R:          1.48
Max Drawdown:      -14.63 %
Sharpe Ratio:       2.20
Total Trades:      5,671
=====
```

Figure 8. Key performance metrics of the final system

Figure 8 summarizes the aggregated risk-adjusted metrics over 5,671 executed trades. The system achieves an average profit per trade of \$2.30, an average of 6.24 trades per active trading day, and

a Profit Factor of 1.14 - indicating that gross profit exceeds gross loss by 14% and representing the primary limitation of the current configuration. The win rate stands at 43.4%, with successful trades yielding an average profit of \$43.88 against an average loss of -\$29.57 on losing trades. The 1.5:1 reward-to-risk asymmetry arises from dynamic early-exit filters that close positions upon detected reversal probability, thereby truncating losses and preserving unrealized gains. Maximum drawdown is contained at 14.63%, the Sharpe ratio reaches 2.20, and the recovery factor of 15.76 underscores strong capital recuperation following adverse periods. All metrics are derived from realistic historical simulations incorporating variable spreads, commissions, and slippage.

10. Monte Carlo Validation

As emphasized in the literature review (Section 2), Monte Carlo simulation represents a critical robustness check for algorithmic trading systems by revealing the distribution of possible outcomes under stochastic resampling of trade sequences. Accordingly, 50,000 independent one-year simulations were conducted using the empirical parameters derived from the historical back test: a win rate of 43.4%, a reward-to-risk ratio of 1.5:1, an initial capital of \$10,000, a fixed per-trade risk of \$50, and an annualized trade count of 1,685 (calibrated to the observed average frequency).

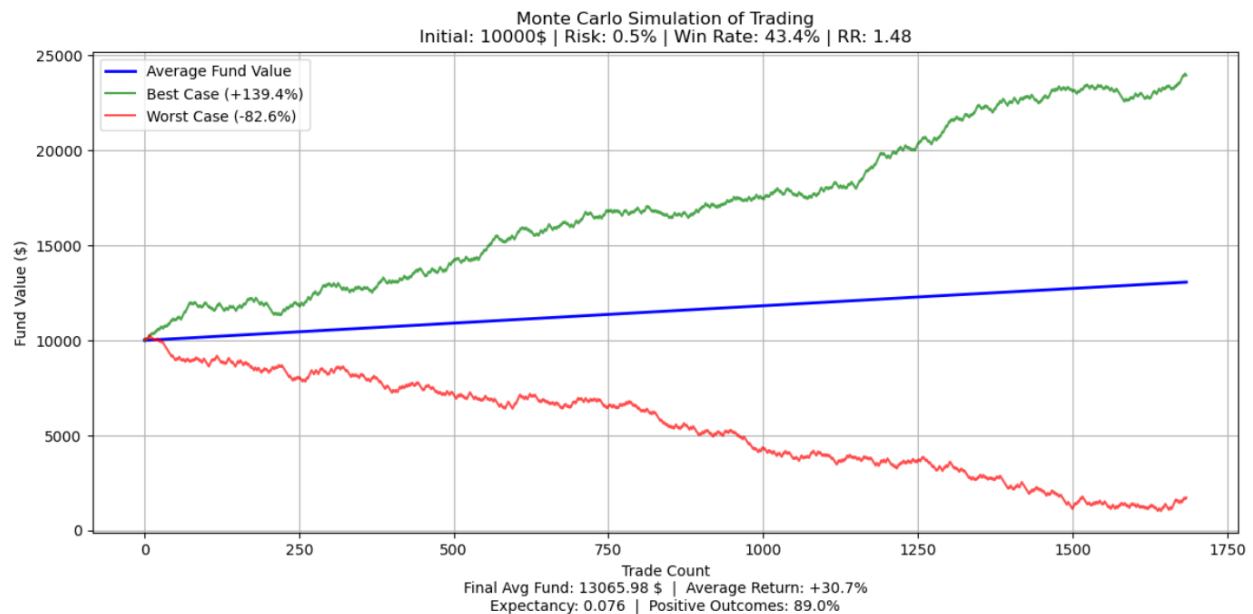


Figure 9. Results of the Monte Carlo simulation (50,000 iterations)

Figure 9 presents the resulting distribution of terminal equity values. The mean annualized return across simulations is 30.7%, with the best-case scenario yielding +139% and the worst-case scenario terminating at -82.6%. The expected profit per trade is estimated at 7.6% of risked capital, and 89% of simulated paths conclude the year with positive net performance.

These findings confirm a strongly positive expectancy and a high probability of annual profitability, consistent with the favorable historical back test. Nevertheless, the substantial tail risk evidenced by occasional large drawdowns underscores the persistent presence of execution, slippage, and regime-shift risks in live deployment. The Monte Carlo analysis thus reinforces the necessity of rigorous risk controls and continuous monitoring when transitioning the system from historical simulation to real-market operation.

11. Conclusion

This study has developed and comprehensively evaluated a hybrid algorithmic trading system for the spot gold market (XAU/USD) that integrates an enhanced XGBoost forecasting engine with an optimized ensemble of rule-based filters and a fixed-dollar-risk position-sizing rule. Utilizing over 300,000 5-minute candlesticks from January 2021 to October 2025, the model achieved strong predictive performance (in-sample MAPE 0.11%, RMSE 3.49; out-of-sample MAPE 0.06%, RMSE 2.83) and, when operationalized as a complete trading strategy, delivered a cumulative net return of 130.62% (\$13,062.38 profit on \$10,000 initial capital) over the five years. Key risk-adjusted metrics included a Profit Factor of 1.14, a win rate of 43.4%, an average winning trade of \$43.88, an average losing trade of -\$29.57 (realized reward-to-risk 1.5:1), a maximum drawdown of 14.63%, a Sharpe ratio of 2.20, and a recovery factor of 15.76 across 5,671 trades. The system demonstrated pronounced regime dependence, exhibiting significantly superior performance during high-volatility sessions (London-New York overlap) and on Mondays, Tuesdays, and Thursdays, while underperforming in low-volume or erratic conditions. Monte Carlo analysis (50,000 iterations) confirmed positive expectancy and an 89% probability of annual profitability, yet highlighted tail-risk vulnerability linked to the modest Profit Factor.

Regarding back-testing limitations, the historical evaluation does not exhibit look-ahead bias or overfitting, since all features and signals are computed strictly from past information and validated out-of-time. However, because the algorithm is designed to execute trades at the exact moment a signal is issued, execution latency becomes an influential factor in live deployment, particularly during periods of elevated market volatility. In live execution, this can occasionally produce order errors, and some trades may fail to activate if the price moves past the intended entry point before the order is registered with the broker.

Regarding computational complexity and generalizability, the framework is not asset-specific in principle and can, in theory, be extended to other instruments. In practice, however, generalization requires asset-by-asset recalibration: the position-sizing relationships, stop-loss/take-profit distances, and model hyperparameters must each be reconfigured and re-optimized separately for every new asset rather than transferred directly.

11.1 Limitations

Despite its robust results, the study exhibits several limitations inherent to single-asset, historical evaluation. First, the system was developed and tested exclusively on spot gold; generalization to other instruments or market regimes remains unverified. Second, transaction costs were modeled conservatively under typical retail broker conditions - higher commissions or wider spreads could materially degrade the realized Profit Factor. Third, no alternative data sources (e.g., news sentiment, order-flow imbalance) were incorporated, potentially constraining performance during macro-driven or sentiment-shift episodes. Finally, while rigorous walk-forward and Monte Carlo procedures were employed, live-market execution risks (latency, partial fills, regulatory changes) were not directly assessed.

11.2 Suggestions for Future Research

Future work could extend the framework in several promising directions. Incorporating large language models for real-time news and social-media sentiment analysis would enable dynamic adjustment of position bias during macroeconomic events. Implementing online or continual learning mechanisms - whereby the XGBoost ensemble incrementally retrains on newly arrived data while preserving stability - offers a pathway toward genuine adaptive behavior. Exploring deep reinforcement learning end-to-end (bypassing explicit forecasting) or transformer-based sequence models may yield further gains in capturing long-range dependencies. Cross-asset and multi-timeframe validation, ensemble combinations with fundamentally different architectures (e.g., LSTM, Temporal Convolutional Networks), and rigorous forward-testing on uncorrelated instruments (major FX pairs, equity indices, cryptocurrencies) would strengthen claims of robustness. Finally, integrating limit-order-book features where available could enhance execution quality in centralized venues.

11.3 Practical Implementation Considerations

Transitioning the system to live trading demands strict operational discipline. The most critical requirement is maintaining the fixed monetary risk per trade (here \$50) irrespective of equity fluctuations; scaling position size proportionally with account equity would amplify drawdowns given the sub-50% win rate and could ultimately destroy long-term expectancy. Users must resist discretionary intervention - frequent activation or deactivation based on short-term results is statistically detrimental. Expected maximum consecutive losses should be precomputed and aligned with individual risk tolerance; exceeding this threshold warrants either full automation cessation or reversion to a decision-support role. Daily profit/loss circuit breakers must remain symmetric: halting only after exceptional single-day gains would allow mean-reversion to erode profits. Continuous monitoring of execution logs is essential to address occasional order rejections under high volatility or connectivity issues; automated retry logic mitigates missed high-conviction opportunities. Finally, the modest Profit Factor (≈ 1.14) renders performance highly sensitive to

transaction costs, so deployment on alternative instruments should be restricted to those exhibiting comparable or superior liquidity and fee structures (e.g., USD/JPY, AUD/CAD). Adherence to these protocols - fixed monetary risk, operational continuity, real-time oversight, and cost compatibility - constitutes the prerequisite for preserving the edge demonstrated in historical testing.

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